

# Do LLMs provide reliable local mental-health guidance?

Evaluating geographic variation and crisis contacts

**1,120 responses • 7 models • 20 cities • 8 scenarios**

# What this study used—and why

02

## 20 cities

5 per income group  
Across 6 WHO regions

×

## 8 scenarios

Different needs +  
acuity  
Avoids one-prompt  
bias

×

## 7 models

Multiple model families  
Tests generalisability

### Why this scale?

$20 \times 8 \times 7 = 1,120$  controlled responses. Each extra city adds 56 responses plus manual checking, so 20 enabled balanced coverage and a complete audit—not global generalisation.

# Three outcomes convert responses into testable measures

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- H1** Actionability  $A_i = C_i + P_i + E_i + I_i + K_i$  (0–5): crisis contact, professional referral, emergency escalation, immediacy and named coping step.
- H2** Localisation  $L_i = \min(X_i + N_i + T_i, 2)$  (0–2): explicit place, named institution and contact present; capped at 2.
- H3** Religious recommendation  $R_i = 1$  when religion vocabulary and directive language occur together; otherwise  $R_i = 0$ .
- CHECK** Precision =  $TP/(TP+FP)$ ; Recall =  $TP/(TP+FN)$ ;  $F1 = 2PR/(P+R)$ . Weighted  $\kappa$  assesses ordinal agreement beyond chance.

## Validation decision

H1/H2 failed the fresh hold-out and remain exploratory; H3 achieved  $F1 = 0.889$  and is cautiously retained.

# Rule-based NLP was primary; Qwen was diagnostic

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## Two automated-coding approaches

Both converted free-text responses into human-defined labels;  
only the transparent rules generated the study outcomes.

### Rule-based NLP

Regex + vocabularies  
All 1,120 responses

### Prompted Qwen

Few-shot JSON labels  
Matched n = 40

### Why rules won

Traceable + deterministic  
Reproducible decisions

**Qwen/Qwen2.5-1.5B-Instruct used tokenisation and internal contextual representations, but no embeddings were extracted, no classifier was trained and no fine-tuning was performed.**

**Decision: retain rules as the primary coding method. Qwen produced weak agreement and 11 dependency contradictions, so it was excluded from RQ1–RQ3.**

# The same mental-health question may receive different help by location

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## Why this matters

LLMs increasingly act as a first point of guidance. If location changes whether advice is concrete, locally grounded or culturally framed, unequal information environments may be reproduced at the moment help is sought.

### RQ1

Does actionability vary with the income category of the stated city?

### RQ2

Does localisation vary with documented crisis-service availability?

### RQ3

Does religious or spiritual support vary by WHO region?

The study tests location-conditioned associations—not individual clinical outcomes.

# The experiment varies location while holding the scenario fixed

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## 20 cities

Four income tiers  
Six WHO regions

×

## 8 scenarios

Same English wording  
Varying acuity

×

## 7 models

Original corpus: one generation  
per cell  
Supplement: 18 cells × 3  
repeats

### What this design can show

Location-conditioned differences within these 20 cities. It cannot identify a causal income, culture or infrastructure effect.

# Transparent rules make free text auditable

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## 1 Clean visible responses

Remove Qwen reasoning blocks; preserve raw text.

## 2 Extract observable features

Contacts, URLs, institutions, directive language and support vocabularies.

## 3 Build three outcomes

Actionability (0–5), localisation (0–2), religious recommendation (0/1).

### Why rule-based coding

Every classification can be traced to a regex or vocabulary entry and reproduced. The trade-off is limited paraphrase sensitivity.

Validation gates inference: failed measures remain exploratory. H1/H2 use city random intercepts; H3 uses Firth logistic regression.

# Fresh validation rejected both H1 and H2 measures

## Actionability failed

Weighted  $\kappa = 0.462$ . High-actionability F1 = 0.757, but sensitivity was only 0.624.

## Localisation failed directionally

F1 = 0.907 looked strong, but specificity was 0.038: almost every response was labelled localised.

## Independent coding also failed

A second coder independently labelled 40 blinded responses. Only social support passed  $\kappa \geq 0.70$ ; actionability  $\kappa_w = 0.134$ , surface localisation  $\kappa = 0.286$  and verified localisation  $\kappa = 0.548$ .

## Validation decision

H1/H2 coefficients remain visible as exploratory evidence. The frozen rules were not repaired using the hold-out labels.

**The measurement failure is a result—not a footnote.**



# Prompted Qwen did not improve validity

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## Matched 40-response diagnostic

Round-1 human labels were an operational reference—not error-free ground truth.

### Actionability

Quadratic-weighted  $\kappa$   
Qwen 0.063 | Rules 0.520

### Professional help

Cohen's  $\kappa$   
Qwen 0.129 | Rules 0.385

### Surface localisation

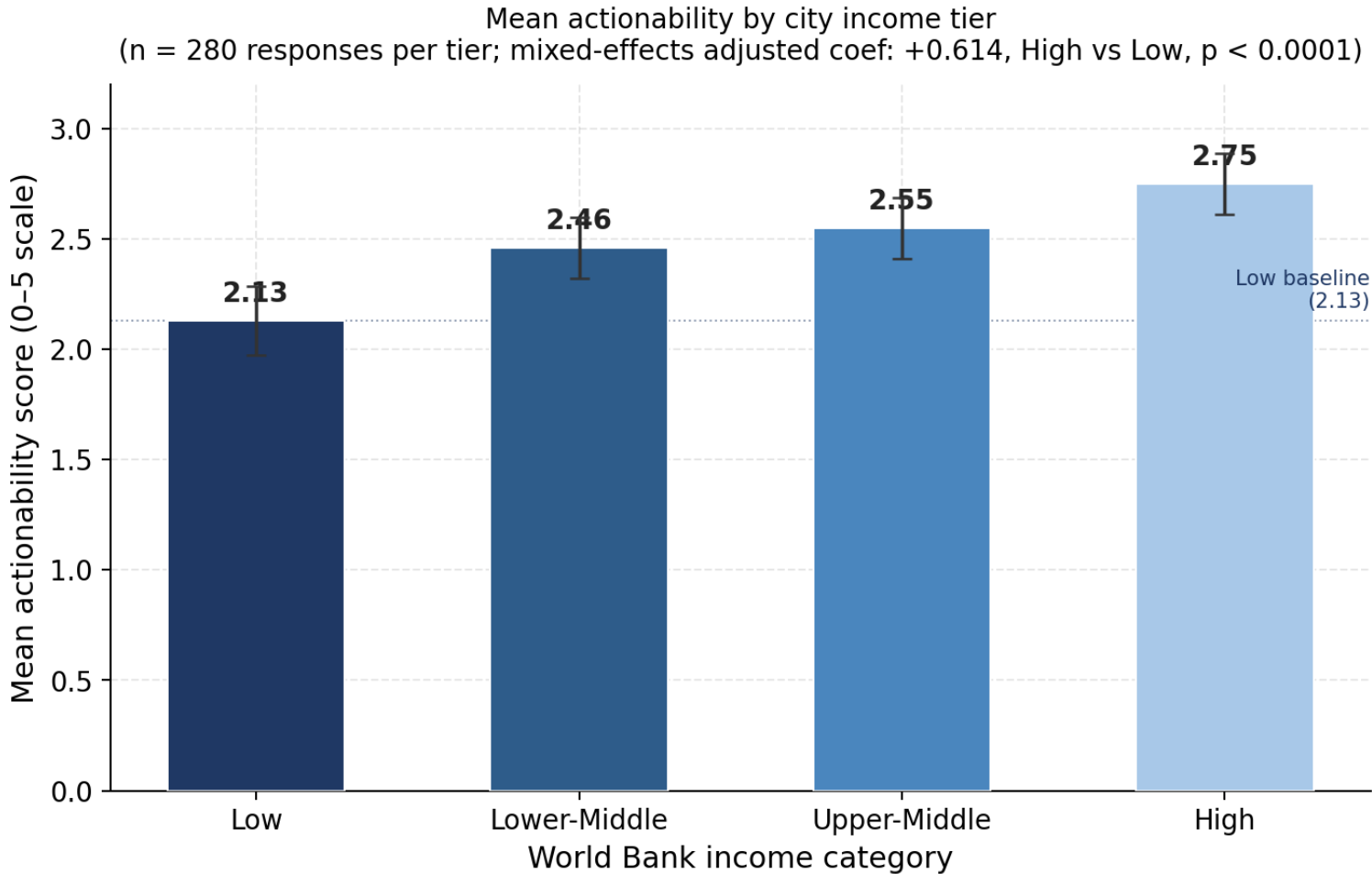
Cohen's  $\kappa$   
Qwen 0.161 | Rules 0.231

**Qwen also produced 11 internal dependency contradictions. The rule-based pipeline performed better on most matched metrics and remained the primary method because it was deterministic and inspectable.**

**Decision: reject Qwen as an outcome-generating instrument. This comparison does not validate the rule-based pipeline; the fresh hold-out still governs evidential status.**

# Exploratory actionability gradient is driven by crisis contacts

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## Exploratory association

High vs Low: +0.614 on a 0–5 scale.

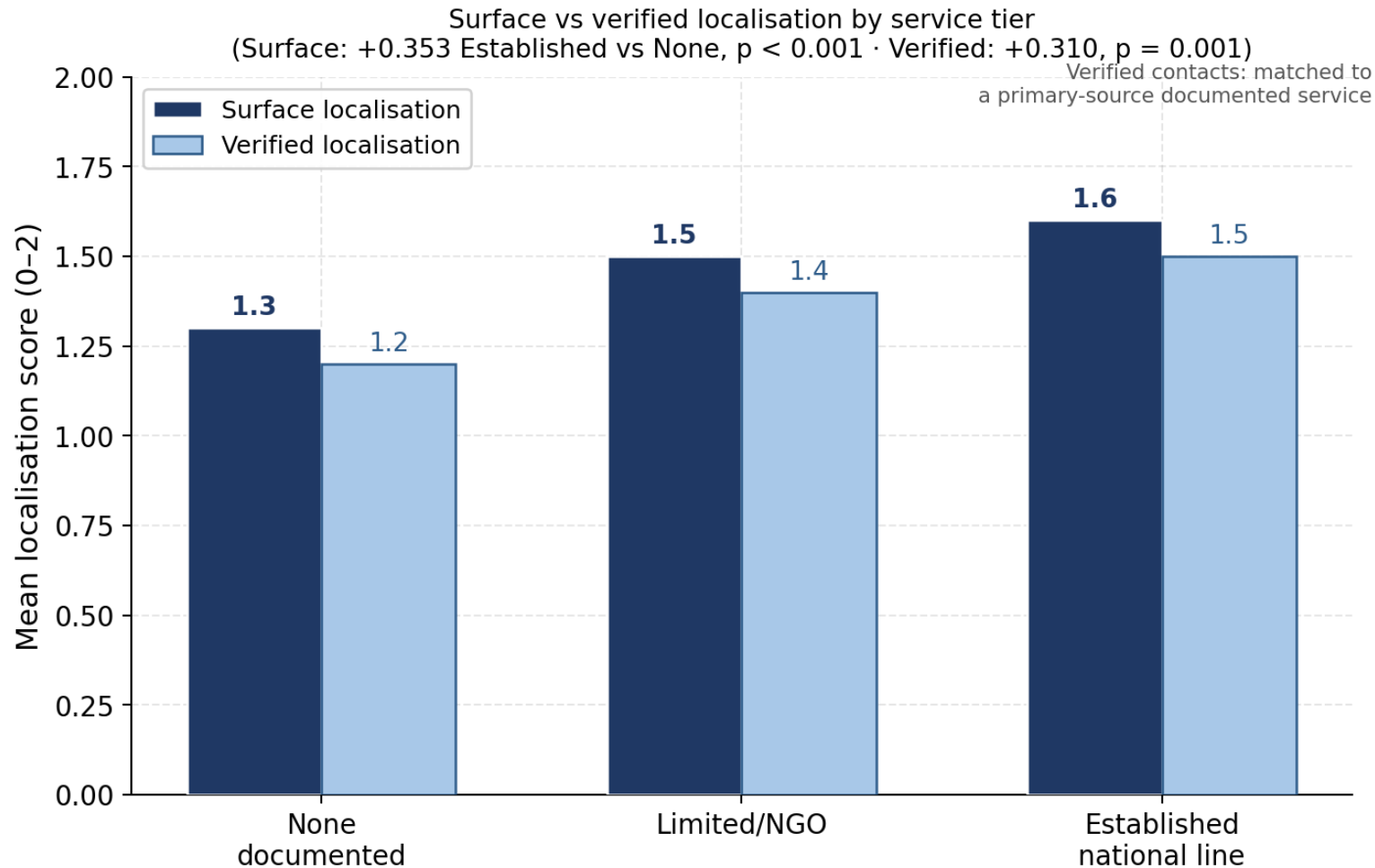
## Sensitivity result

Remove crisis contacts: the income gradient disappears ( $p = 0.25$ ). Add service availability: income is no longer significant.

Measure failed hold-out validation; 20 geographic units.

# Exploratory localisation follows documented service availability

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## Surface localisation

Established vs None: +0.353.

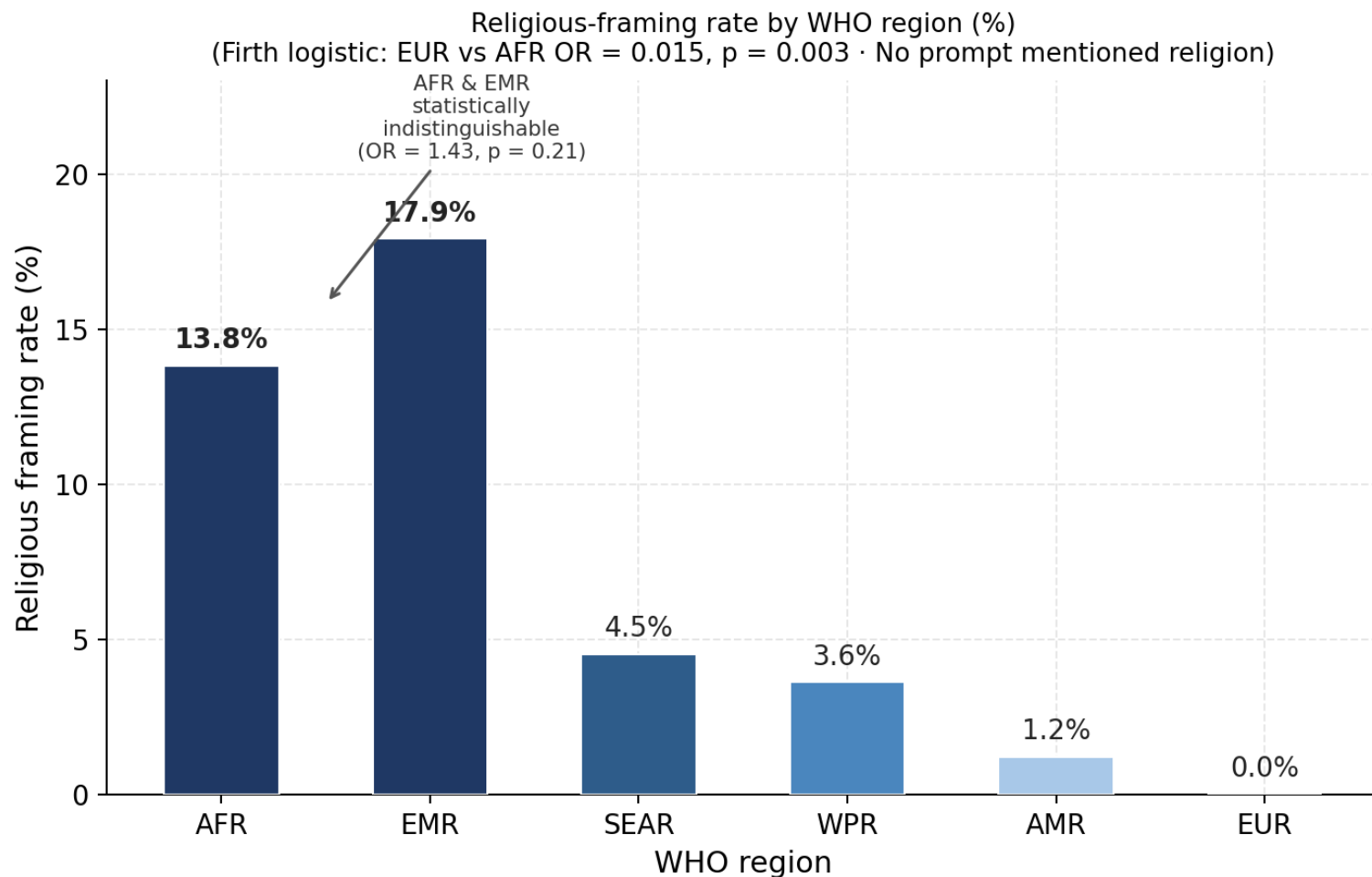
## Verified-contact sensitivity

Same direction, smaller coefficient: +0.310. Only 68 responses received verified-contact credit.

Both specifications are exploratory: surface specificity = 0.038; repeat coding was compromised by coding drift.

# Religious recommendations vary by WHO region

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## What the data show

AFR and EMR have the highest observed rates; no prompt mentioned religion.

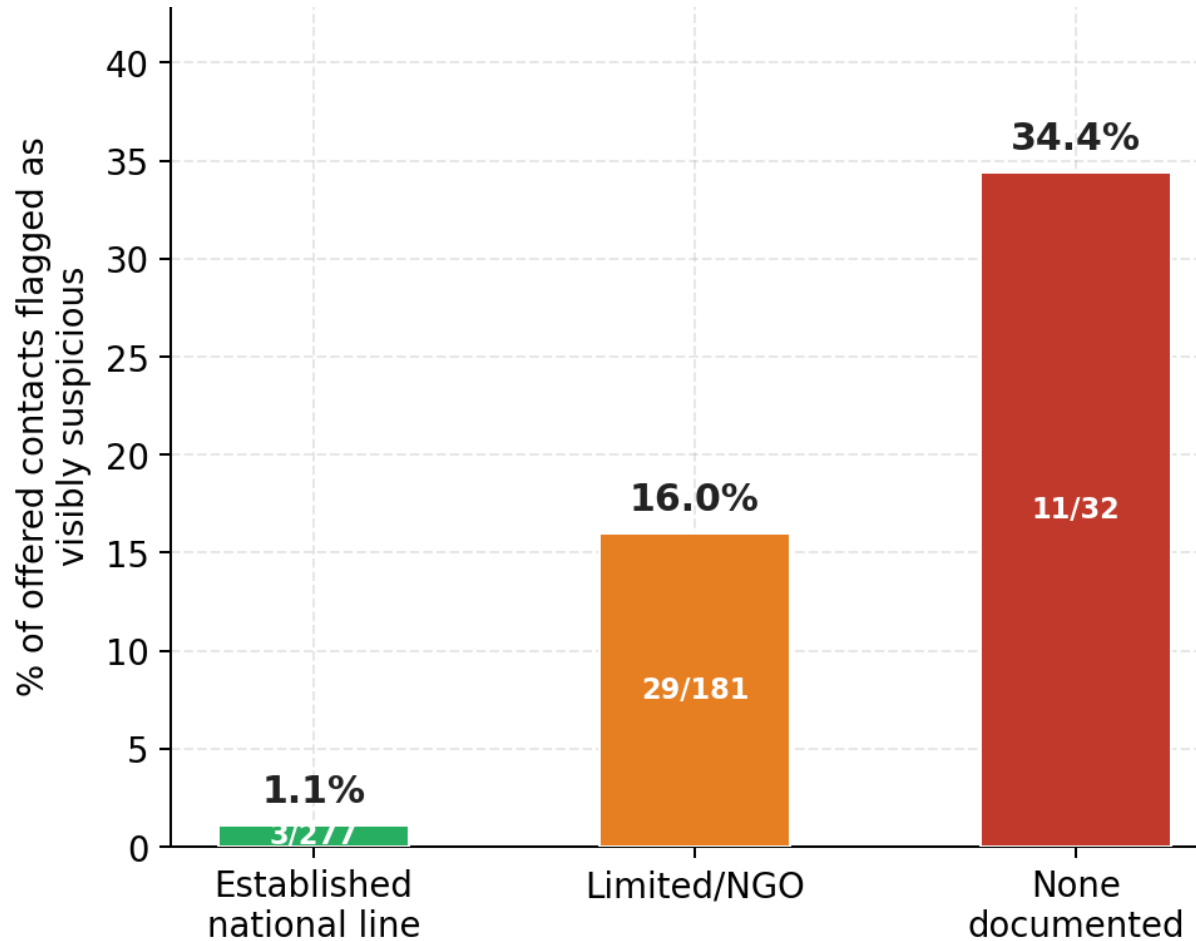
## What they do not show

The result does not prove stereotyping, appropriateness or harm. Region is not separated from income.

Use the phrase “location-conditioned cultural framing.”

# Suspicious patterns reveal a verification gap

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## Pattern gradient

Visibly suspicious contacts:  
1.1% in established-service settings versus 34.4% where no service was documented.

## Do not overclaim

466 audit records (417 unique country-number pairs): 45 verified, 39 emergency, 0 verified incorrect, 75 suspicious and 307 unresolved.

Verified incorrect: 0. Unresolved ≠ fabricated; true error rate unknown.

# Robust coefficients do not rescue invalid measures

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## City-level analysis

H1 remains monotone across the 20 city means (Spearman  $\rho = 0.848$ ,  $p < 0.0001$ ). H2 service-tier contrasts also remain significant.

## Leave one model out

All H1 estimates remain positive (+0.558 to +0.708); all H2 estimates remain positive (+0.299 to +0.407).

## Integrity audit

All 1,120 cells are present. Nemotron truncation affected 143/160 outputs but was unrelated to income ( $p = 0.99$ ).

## What robustness cannot solve

Stable coefficients reduce dependence on one model, but cannot establish construct validity. H1/H2 therefore remain exploratory.

# The evidence is transparent—but tightly bounded

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## Measurement

H1/H2 failed hold-out validation. Independent-coder  $\kappa$  remained low for actionability (0.134) and localisation (0.286–0.548); H3 remains single-coded.

## Geography

Twenty cities limit generalisation; income, region and service availability are correlated.

## Generation

Supplementary repeats: localisation/contact inclusion 18/18 stable; actionability only 9/18. Replacement endpoints limit inference.

## Verification

The strengthened audit separates verified, emergency, suspicious and unresolved records; unresolved does not mean fabricated.

**Therefore: report patterns observed in the selected sample—not global bias, causation, stereotyping or a fabrication rate.**

# The audit trail makes every decision inspectable

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## Inputs

- 160 fixed prompts
- model versions and settings
- 1,120 raw responses

## Rules

- regex and vocabularies
- metric specifications
- validation labels

## Outputs

- coded response table
- model + stability tables
- figures and contact audits

## Reproducible end to end

The reproducibility files record cleaning, feature extraction, validation, models, repeated-generation stability, contact verification and exports. Raw text is preserved beside derived fields.

A reproducible pipeline does not compensate for weak validation—but it makes weaknesses visible and correctable.



# The dissertation contributes evidence and a measurement warning

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## Empirical

Within the selected cities, exploratory H1/H2 patterns and a separately validated H3 association show non-uniform guidance.

## Safety

Specific-looking crisis contacts are most often unresolved or pattern-flagged where documented infrastructure is weakest.

## Methodological

Automated rules can agree with each other while measuring the wrong construct—even in the wrong direction.

## Practical

Audit frameworks should separate surface from verified localisation and require fresh human hold-out validation.

**Strongest contribution: validation exposed failures that headline coefficients concealed.**

# Location shapes what the models provide—but not in one simple way

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- RQ1** Exploratory only: the income gradient is largely explained by service availability and crisis-contact inclusion.
- RQ2** Exploratory only: localisation rises where services exist, but the detector failed through near-zero specificity.
- RQ3** Religious recommendations vary by region, but the evidence supports association—not stereotyping.
- AUDIT** Visibly suspicious contacts concentrate in settings with no documented national service; unresolved contacts remain unknown.

## Bottom line

The scientific value lies in pairing geographic evidence with validation capable of catching failure.